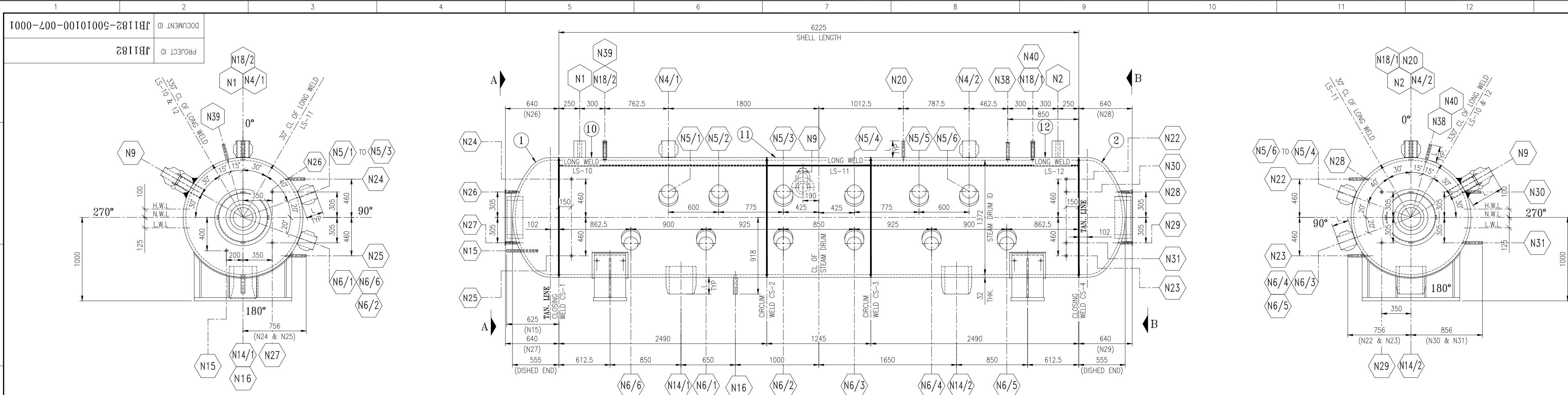
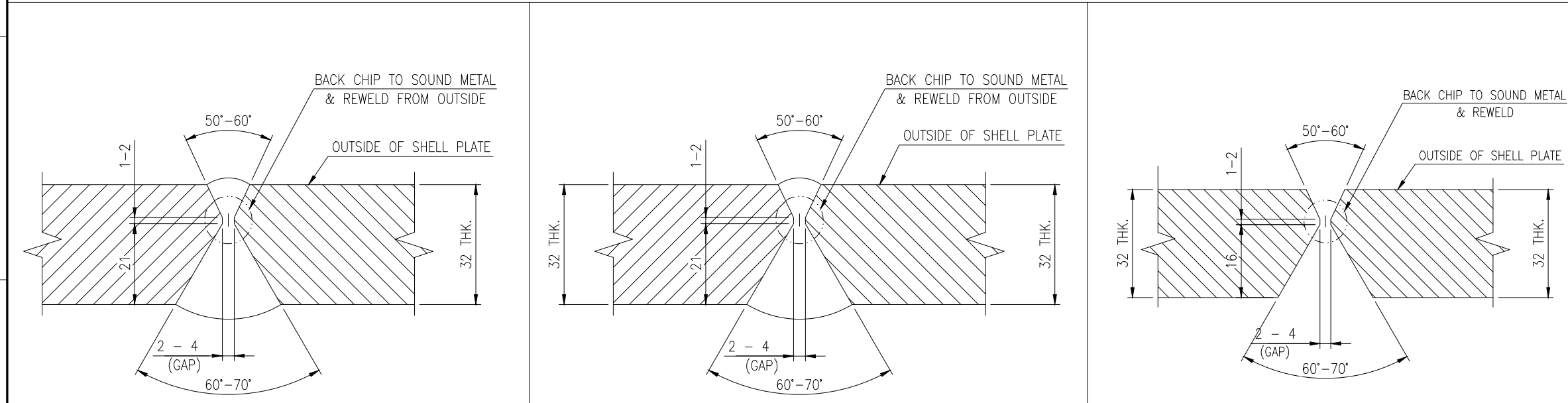




VIEW - AA
(LHS OF BOILER)

ELEVATION (VIEWED FROM FRONT OF BOILER)
ARRANGEMENT OF STEAM DRUM

VIEW - BB
(RHS OF BOILER)

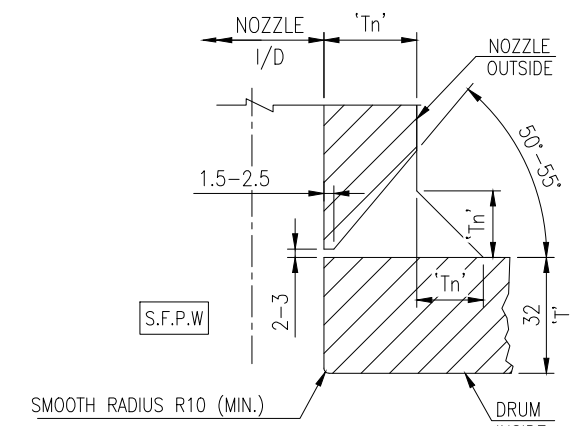


WELD PREPARATION OF LONG SEAMS FOR STEAM DRUM
(SEAM MK. Nos. LS-10, LS-11 & LS-12)

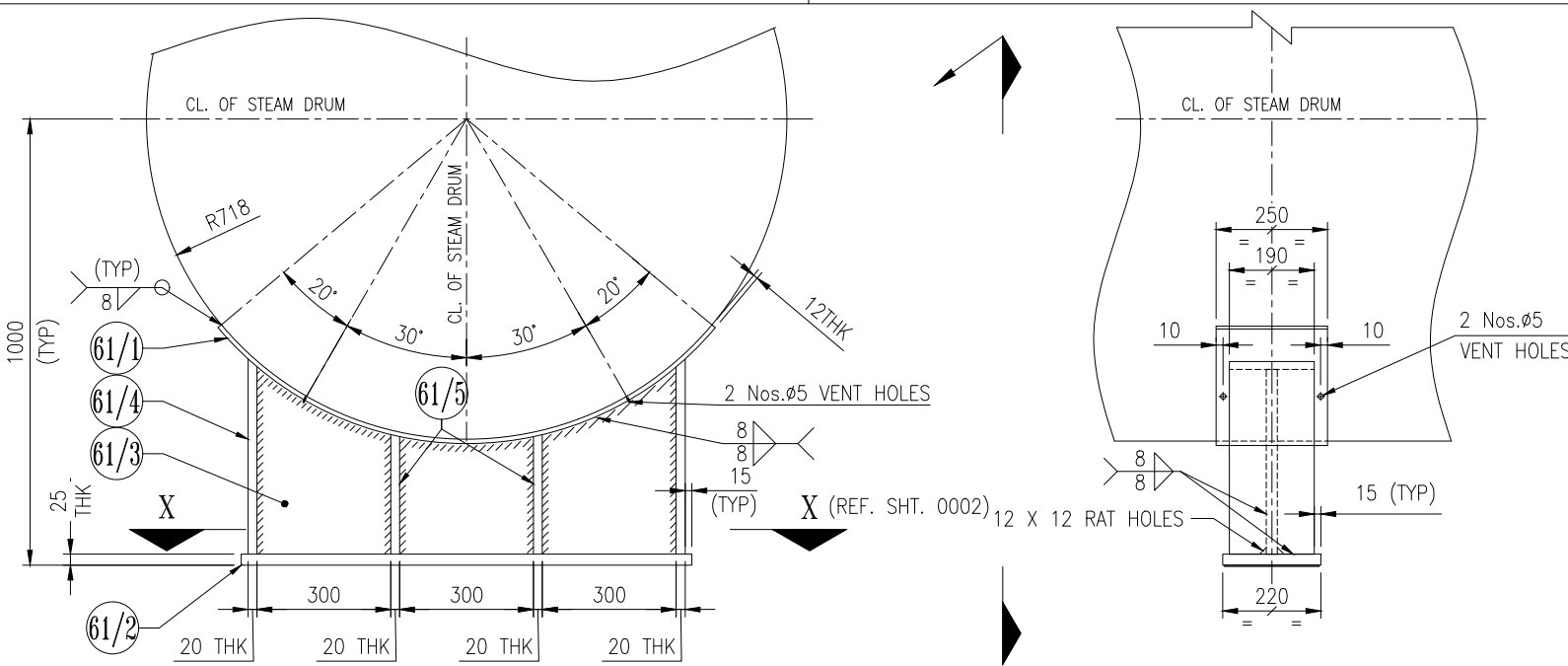
WELD PREPARATION OF CIRCUMFERENTIAL SEAMS FOR STEAM DRUM
(SEAM MK. Nos. CS-2 & CS-3)

WELD PREPARATION OF CLOSING SEAM FOR STEAM DRUM
FOR MARK Nos. CS-1 & CS-4

IBR CHAPTER - XII
IBR FIG : XII/85E B2

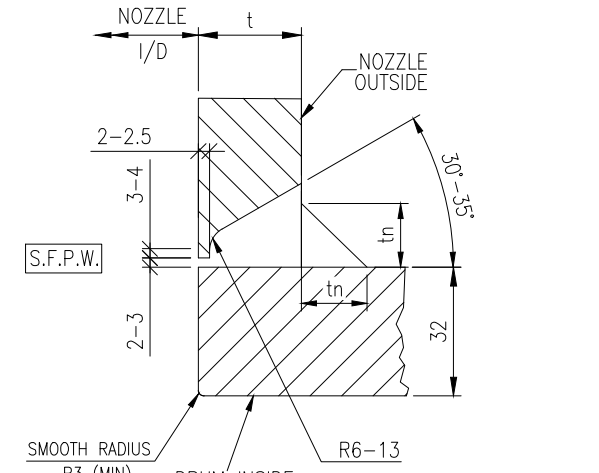


[FOR MARK Nos. N1, N2]
WELDING DETAIL OF NOZZLE TO DRUM



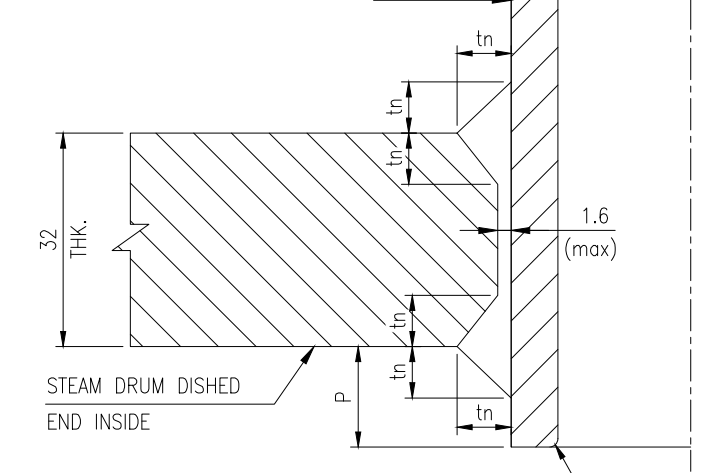
[ALL WELDS ARE 0.71 mm FILLET WELD]
DETAIL OF STEAM DRUM TEMPORARY SUPPORT

IBR CHAPTER - XII
IBR FIG : XII/85E J2



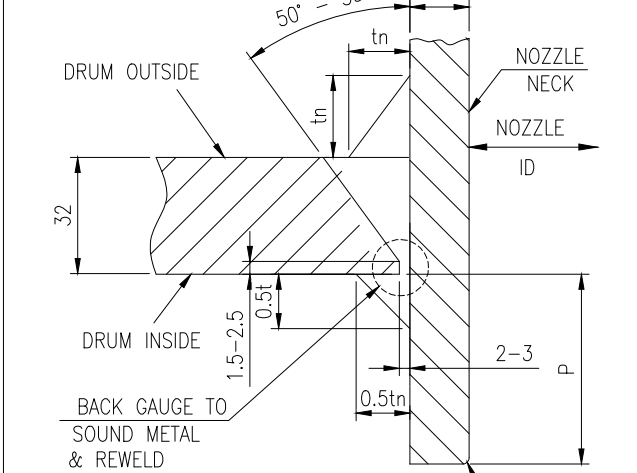
WELD DETAIL OF NOZZLE TO DRUM
[FOR MARK Nos. N4/1 & N4/2, N5/1 TO N5/6, N6/1 TO N6/6, N9, N16, N18/1, N18/2, N20, N22, N23, N24, N25, N26, N27, N28, N29, N30, N31, N38, N39, N40]

CHAPTER-VIII
IBR FIG : 365/13

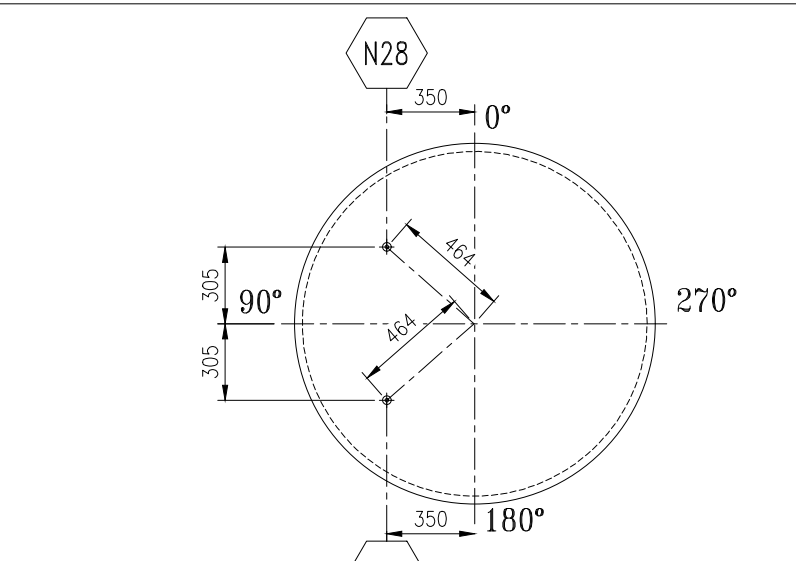


[FOR MARK Nos. N15]
WELDING DETAIL OF NOZZLE TO DRUM

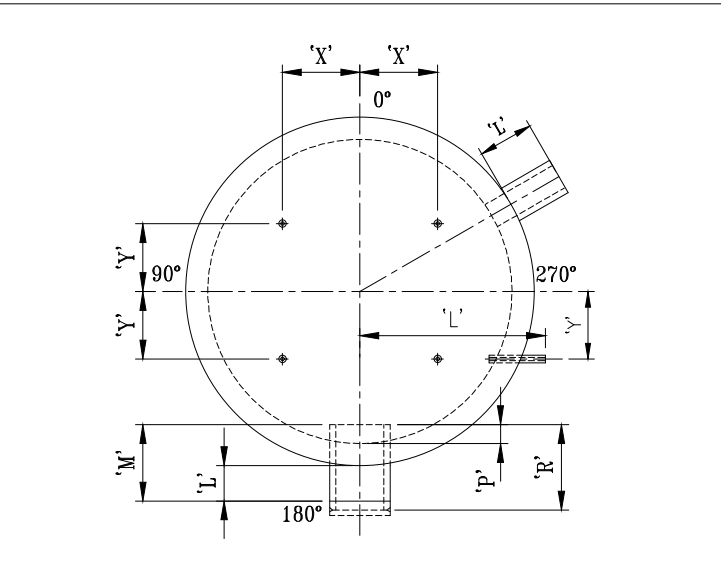
CHAPTER - XII
IBR FIG : XII/85E B1



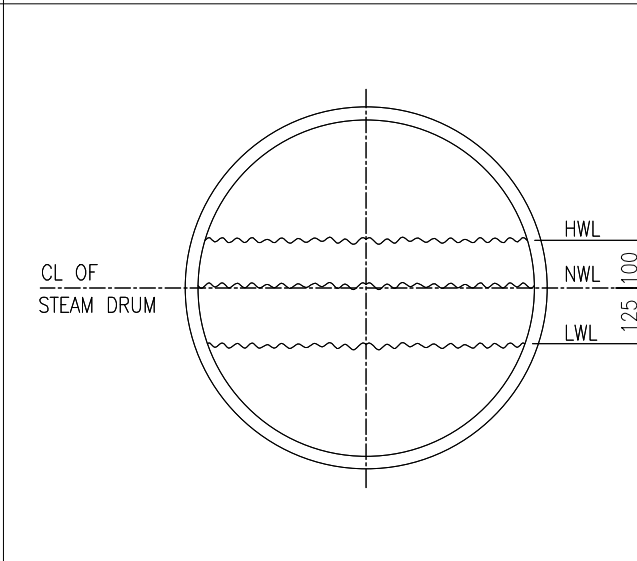
WELD DETAIL OF NOZZLE TO DRUM
MARK No. N14/1 & N14/2



DETAIL OF STEAM DRUM DISHED END
PIPE NOZZLES
(TYP. FOR RHS)



TYP. DIM. IDENTIFICATION VIEW



WATER LEVELS FOR STEAM DRUM

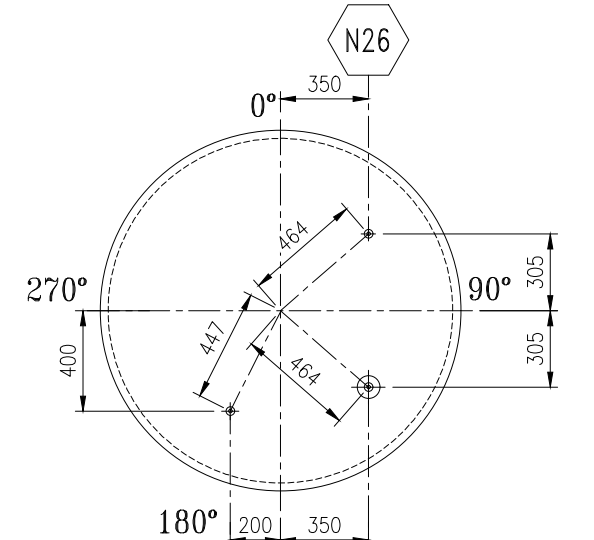
DESIGN DATA	
CODE OF CONSTRUCTION	I.B.R. (1950) AND IT'S AMENDMENTS UPTO SEPTEMBER 2020 (SEE NOTE : 02)
MAXIMUM WORKING PRESSURE	59 Kg/Sq. cm (g)
MAXIMUM WORKING TEMPERATURE	275 °C
WORKING METAL TEMPERATURE	275 °C
HYDRAULIC TEST PRESSURE	88.5 Kg/Sq. cm (g) (AT SHOP & AS PER IBR AT SITE)
RADIOGRAPHY OF LONG. & CIRCUM. SEAM	100%
STRESS RELIEVING	YES (REFER. HEAT TREATMENT CYCLE TABLE)
TOTAL HEATING SURFACE AREA OF BOILER (EXCLUDING ECONOMISER ELEMENT & SUPER HEATER)	1700 Sq. m
HEATING SURFACE AREA OF ECONOMISER SECTION (NON STEAMING)	1230 Sq. m
STEAM PRESSURE AT SUPER HEATER OUTLET	45 Kg/Sq. cm (g)
SUPER HEATER OUTLET TEMPERATURE	400±10 °C
WEIGHT OF DRUMS (EMPTY)	10359.2 Kg. (STEAM DRUM)
INSPECTION	DIRECTOR OF BOILERS, UTTAR PRADESH/Lloyds (SEE NOTE : 04 IN 007-0002)

LIST OF REFERENCE DRAWINGS	
DRAWING DOCUMENT ID	DESCRIPTION
JB1182-50010100-006-0001	ARRANGEMENT & DETAIL OF CIRCULAR MANHOLE ON STEAM DRUM
JB1182-50010100-007-0002	DETAILS & MATERIAL LIST FOR STEAM DRUM
JB1182-50010100-009-0001	DETAIL OF DRUM NOZZLES

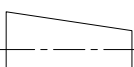
MATERIAL SPECIFICATIONS :					
DESCRIPTION	SIZE	SPECIFICATION	MIN. TENSILITY (Kg/ Sq.mm)	MIN. CALC. THK. (mm)	REMARKS
SHELL BELTS	AS PER MATERIAL LIST	SA 516 Gr.70	49.45	30.36	FOR STEAM DRUM
DISHED ENDS	AS PER MATERIAL LIST	SA 516 Gr.70	49.45	25.01	FOR STEAM DRUM

- NOTES :-**
01. ALL DIMENSIONS ARE IN "mm" UNLESS OTHERWISE SPECIFIED.
 02. THE ZERO DATE OF THIS CONTRACT IS 17.09.2020 . HENCE IBR AMENDMENTS UP TO SEPTEMBER 2020 HAVE BEEN TAKEN CARE OF WHILE DESIGNING THE EQUIPMENT.
 03. INSPECTION BY DIRECTOR OF BOILERS, UTTAR PRADESH/Lloyds CBB Authorisation, Form XMA See Regulation 40(2), BECAUSE MANUFACTURING OF THIS EQUIPMENT SHALL WILL BE DONE AT ATV PROJECT INDIA LIMITED, MATHURA, UTTAR PRADESH.
 04. FOR GENERAL NOTES & MATERIAL SPECIFICATIONS REF. DOCUMENT ID : JB1182-50010100-007-0002.
 05. PWHT CYCLE :- STEAM DRUM
 - A) LOADING TEMP. (MAX.) = 300°C
 - B) RATE OF HEATING ABOVE 300°C (MAX.) = 100°C/Hr.
 - C) SOAKING TEMP. = 610°C ± 10°C
 - D) SOAKING TIME (MIN.) = 1 Hr 20 MINUTES.
 - E) RATE OF COOLING ABOVE 300°C (MAX.) = 100°C/Hr.
 - F) UNLOADING TEMP (MAX.) = 300°C
 - G) RATE OF HEATING AFTER UNLOADING = COOLED IN STILL AIR

S.No.	Nozzle Nos.	Nozzle Descriptions	Location of Nozzle	No. of Nozzle	Nozzle Size (OD) (D) mm	Schedule of Nozzle	Nozzle Thickness (t) mm	Connecting Pipe Thickness (mm)	Projection from Drum Vertical Axis (mm)	Drum Inside Bolt Circle Dia (mm)	Nozzle CL to Drum Vertical Axis (Z) mm	Nozzle CL to Drum Horizontal Axis (X) mm	Calculated Nozzle Length (B) mm	Finished Length of the Nozzle (M) mm	M/C. Back End Preparation (E) mm	Weld Dim. (In) mm	Nozzle Inside Projection (P) mm
1	N1	SAFETY VALVE NOZZLE-1	STEAM DRUM SHELL	1	↑	↑	—	—	—	—	—	—	—	—	—	—	—
2	N2	SAFETY VALVE NOZZLE-2	STEAM DRUM SHELL	1	↑	↑	—	—	—	—	—	—	—	—	—	—	—
3	N4/1 & N4/2	SATURATED STEAM STUB	STEAM DRUM SHELL	2	230	—	37.96	SCH.40	200	—	—	—	250	200	154.08	32	0
4	N5/1 & N5/2	SIDE WALL RISER STUB	STEAM DRUM SHELL	6	230	—	37.96	SCH.40	200	—	—	—	250	200	154.08	32	0
5	N6/1 to N6/6	FRONT, REAR WALL & EVAPORATOR RISER STUB	STEAM DRUM SHELL	6	230	—	37.96	SCH.40	200	—	—	—	250	200	154.08	32	0
6	N9	FEED INLET (STUB + THERMAL SLEEVE)	STEAM DRUM SHELL	1	215	—	40.00	—	—	—	—	—	REF. DRG. -009-0001	—	—	—	125
7	N14/1 & N14/2	COMMON FEEDER NOZZLE	STEAM DRUM SHELL	2	360	—	39.49	SCH.100	200	—	—	—	385	332	281.02	32	100
8	N15	CRD NOZZLE	STEAM DRUM DISHED END	1	33.4	SCH.XXS	9.09	SCH.40	—	200	400	420	370	26.64	9	151	—
9	N16	CRD NOZZLE	STEAM DRUM SHELL	1	48.3	SCH.XXS	10.15	SCH.40	200	—	—	—	250	200	40.94	10	0
10	N18/1 & N18/2	AIR VENT	STEAM DRUM SHELL	1	48.3	SCH.XXS	10.15	SCH.40	200	—	—	—	250	200	40.94	10	0
11	N20	INSTRUMENT TEST GAUGE NOZZLE	STEAM DRUM SHELL	1	33.4	SCH.XXS	9.09	SCH.40	200	—	—	—	250	200	26.64	9	0
12	N22	LEVEL GAUGE NOZZLE (STEAM)	STEAM DRUM SHELL	1	33.4	SCH.XXS	9.09	SCH.40	—	460	270	219	26.64	9	0	—	—
13	N23	LEVEL GAUGE NOZZLE (WATER)	STEAM DRUM SHELL	1	33.4	SCH.XXS	9.09	SCH.40	—	460	270	219	26.64	9	0	—	—
14	N24	LEVEL GAUGE NOZZLE (STEAM)	STEAM DRUM SHELL	1	33.4	SCH.XXS	9.09	SCH.40	—	460	270	219	26.64	9	0	—	—
15	N25	LEVEL GAUGE NOZZLE (WATER)	STEAM DRUM SHELL	1	33.4	SCH.XXS	9.09	SCH.40	—	460	270	219	26.64	9	0	—	—
16	N26	LEVEL TRANSMITTER (STEAM)	STEAM DRUM DISHED END	1	33.4	SCH.XXS	9.09	SCH.40	—	350	305	270	216.5	26.64	9	0	—
17	N27	LEVEL TRANSMITTER (WATER)	STEAM DRUM DISHED END	1	33.4	SCH.XXS	9.09	SCH.40	—	350	305	270	216.5	26.64	9	0	—
18	N28	LEVEL TRANSMITTER (STEAM)	STEAM DRUM DISHED END	1	33.4	SCH.XXS	9.09	SCH.40	—	350	305	270	216.5	26.64	9	0	—
19	N29	LEVEL TRANSMITTER (WATER)	STEAM DRUM DISHED END	1	33.4	SCH.XXS	9.09	SCH.40	—	350	305	270	216.5	26.64	9	0	—
20	N30	LEVEL TRANSMITTER (STEAM)	STEAM DRUM SHELL	1	33.4	SCH.XXS	9.09	SCH.40	—	305	265	214	26.64	9	0	—	—
21	N31	LEVEL TRANSMITTER (WATER)	STEAM DRUM SHELL	1	33.4	SCH.XXS	9.09	SCH.40	—	305	265	214	26.64	9	0	—	—
22	N38	PRESSURE GAUGE NOZZLE	STEAM DRUM SHELL	1	33.4	SCH.XXS	9.09	SCH.40	200	—	—	—	250	200	26.64	9	0
23	N39	PRESSURE TRANSMITTER NOZZLE	STEAM DRUM SHELL	1	33.4	SCH.XXS	9.09	SCH.40	200	—	—	—	250	200	26.64	9	0
24	N40	REMOTE PRESSURE GAUGE NOZZLE	STEAM DRUM SHELL	1	33.4	SCH.XXS	9.09	SCH.40	200	—	—	—	250	200	26.64	9	0



DETAIL OF STEAM DRUM DISHED END
PIPE NOZZLES
(TYP. FOR LHS)

I.W.T. No : 6681											
CONSULTANT		--									
CLIENT		M/s. RANA SUGARS LIMITED, BELWARA (UP)									
DO NOT SCALE IF IN DOUBT PLEASE ASK			THIS DESIGN IS THE PROPERTY OF M/S. ISGEC HEAVY ENGINEERING LTD. AND IS ONLY ALLOWED TO BE USED BY EXPRESSED PERMISSION AND LICENSE FROM M/S. ISGEC HEAVY ENGINEERING LTD.								
FOR APPROVAL		NO	FOR INFORMATION		NO	FOR PRODUCTION		YES	FOR ERECTION		YES
RESPONSIBLE DEPT.		DRAWN BY		CHECKED BY		APPROVED BY		DATE			
MECHANICAL		V.KAVASKAR		R.G.AMIR/R.KUMARAN		R.ARUNKUMAR		21.10.2020			
				TOTAL WEIGHT IN kg.		PROJECT ID		ELEMENT ID		I.W.T. NO.	
		REF-007-0002		JB1182		50010100		6681			
YEAR : 2020				SHEET SIZE : A1				SCALE : 1 : 30			
				SLOP FIRED TG BOILER GRTH SUPPORTED							
(formerly Isgec John Thompson)				ARRANGEMENT AND DETAIL OF STEAM DRUM							
Isgec Heavy Engineering Ltd. Noids 201 301 (U.P.) India				DRAWING DOCUMENT ID				JB1182-50010100-007-0001			
								REVISION		00	



ISGEC
(formerly Isgec John Thompson)
Isgec Heavy Engineering Ltd.
Noida 201 301 (U.P.) India

REVN.	ZONE	DESCRIPTION	REVN.	ZONE	DESCRIPTION	REVN.	ZONE	DESCRIPTION	REVN.	ZONE	DESCRIPTION
1			2			3			4		
5			6			7			8		
9			10			11			12		
13			14			15			16		